3.6.3 Magnetism and electromagnetism

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 5			
The poles of a magnet are the places where the magnetic forces are strongest. When two magnets are brought close together they exert a force on each other. Two like poles repel each other. Two unlike poles attract each other. Attraction and repulsion between two magnetic poles are examples of non-contact force. The patterns of magnetic fields between bar magnets will be required.		6.7.1.1	4.6.3.1
Outcome 6			
When a current flows through a conducting wire a magnetic field is produced around the wire. The strength of the magnetic field depends on the current through the wire and the distance from the wire. Shaping a wire to form a solenoid increases the strength of the magnetic field created by a current through the wire. Adding an iron core increases the magnetic field strength of a solenoid. An electromagnet is a solenoid with an iron core. Students should be familiar with common uses of electromagnets, eg in scrapyard cranes and relays.	**Suggested activity for TDA** Investigate factors that affect the strength of an electromagnet.	6.7.2.1	4.6.3.4